

Tools and Techniques of Bioinformatics (QUIZ)

Section: 12

Marks: 20

Time: 30 minutes

Answer any 10 questions. Each question is worth 1 points.

For questions that include accession numbers, use web tools to retrieve the necessary information.

Good luck!

* Indicates required question

1. Email *

2. Name *

3. How many "C" nucleotides are present in the *Aplysia californica* insulin mRNA (Accession number NM_001204686.1)?

1 point

Mark only one oval.

☐ 968

☐ 245

☐ 255

☐ 244

4. What is the calculated GC content of the Guinea pig *insulin* gene, complete cds (Accession number K02233.1)? 1 point

Mark only one oval.

- ☐ 67
☐ 75
☐ 57
☐ 25

5. How many histidine (H) amino acids are present in the **NiaP** protein of *Arabidopsis thaliana* (Accession number OAP05651.1)? 1 point

Mark only one oval.

- ☐ 6
☐ 9
☐ 22
☐ 5

6. What is the theoretical PI and molecular weight of **PreM** protein of *Usutu virus* (Accession number YP_164809.1) 1 point

Mark only one oval.

- ☐ 14581 & 69
☐ 77 & 18745
☐ 14581 & 9.19
☐ 8.75 & 18983

7. What is the popular BLAST searching matrix?

1 point

Mark only one oval.

- ☐ BLOSUM65
- ☐ BLOSUM62
- ☐ PAM30
- ☐ BLOSUM55

8. What is the calculated virulence factor for the nephew of atonal protein *Saccoglossus kowalevskii* (Accession number NP_001158470.1)?

1 point

Mark only one oval.

- ☐ 14
- ☐ 2
- ☐ 0
- ☐ None of the above

9. Given the following mRNA sequence:

1 point

"TCAGTTCCACGGTCGTGCCAGA"

What is the dot-bracket form of this sequence?

Mark only one oval.

- ☐ ..(((((((.((((.....
- ☐(((...)))...
- ☐ ...((...))...
- ☐ The sequence does not form a secondary structure.

10. What is the melting temperature of the following primer
"AATTATGCCAATAATATGAACTAT" ?

1 point

Mark only one oval.

- ☐ 58
☐ 72
☐ 41
☐ 74

11. what is the ΔG value of the given sequence "GATTATGCCAATAAGGTGA"? 1 point

Mark only one oval.

- ☐ 90
☐ -1.90
☐ -0.90
☐ -0.84

12. What is the GC content of the primer "GCACGTAGCATGTCACATGCTG"? 1 point

Mark only one oval.

- ☐ 60%
☐ 54.5
☐ 41.5
☐ 50

12:29 PM

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13. What do you think are the possible essential applications or contributions of bioinformatics in the field of agriculture? *

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